

Selection Fronts at an Admissibility Boundary:

Local Exit Geometry, Conditional Boundary Layers, and Reset Semantics

Peter Nero

Version 2, July 2026

Abstract

An admissibility boundary is the place where at least one declared control condition loses positive reserve. That static fact does not by itself produce a boundary layer, metastability, a large-deviation law, a measurement outcome, or a new physical phase. Those conclusions require a specified dynamical model.

This paper gives a precise conditional meaning to the term *selection front*. The input is a finite vector of normalized signed reserve rows, including the rows needed for spectral separation, projector control, well-posedness, descent, coherent stability, truncation, and numerical certification. The exact boundary is distinguished from a conservative numerical event guard. At one regular active row the boundary is a smooth facet; at transverse row ties it is locally an orthant corner. A dynamical front additionally requires a drift, diffusion, initial law, stopping rule, interior return set, observables, and a post-exit stop, chart-transition, reset map, or reset kernel.

The central calculation is an exactly solvable normal diffusion on an interval. Its guard-hitting committor and mean exit time are derived in closed form. Drift away from the guard gives an exponential boundary layer of thickness proportional to noise divided by drift; zero drift gives a linear committor and inverse-noise exit time; drift toward the guard gives neither behavior. Thus no universal sigmoid, metastable law, or first-passage scaling follows from admissibility alone. Eyring–Kramers and Freidlin–Wentzell asymptotics remain available when a small-noise family, attracting basin, quasipotential barrier, and regularity hypotheses are independently supplied.

The result is a reproducible local theory of first exit and a strict application contract. It does not derive vacuum selection, black-hole dynamics, cosmological measures, quantum collapse, or limits on scalable coherence.

Revision note: Version 2

Supersedes Version 2 supersedes Version 1.0, DOI 10.5281/zenodo.18262389.

Reason Version 1 used one unspecified scalar as a universal distance to failure and then asserted necessary boundary layers, metastability, large deviations, abrupt effective-theory failure, measurement collapse, black-hole information loss, cosmological measure failure, and coherence limits. It supplied neither a complete reserve ledger nor the generator, small-noise family, guard, reset law, asymptotic regime, or error certificate needed for those deductions.

Resolution The revision starts from the normalized multirow admissibility record developed in the corrected capacity papers. It defines local front geometry, separates exact and numerical guards, specifies the additional data required for a diffusion or hybrid process, derives an exact one-dimensional exit model, and states the precise extra assumptions under which large-deviation or Eyring–Kramers results may be imported.

Retained content

The useful idea survives: when a selected reduced model approaches the first failure of one of its own control rows, committors, exit times, active-row statistics, and exit-location laws can reveal a sharply localized transition layer.

Remaining boundary

Modal Triplet Theory does not currently select a universal drift, diffusion, noise scale, potential, observation protocol, or post-exit reset for all physical applications. The phrase “selection front” names a conditional first-exit construction, not an additional fundamental interaction.

Contents

Revision note: Version 2	1
1 The question in its corrected form	3
1.1 What is not claimed	3
2 The complete static input	4
2.1 A vector of signed normalized reserves	4
2.2 Exact boundary, front layer, and active rows	4
2.3 Exact and numerical guards are not the same	5
3 Local geometry: facets, corners, and singular points	5
4 The additional dynamical data	6
4.1 A typed front record	6
4.2 Committors and stopping times	7
4.3 A diagnostic barrier becomes a force only when inserted	7
4.4 Exit is not continuation	7
5 An exactly solvable normal front	7
5.1 Model	8
5.2 A genuine boundary layer, under one sign of drift	8
5.3 What this model does and does not establish	9
6 When large deviations and metastability are valid	9
6.1 Large deviations require a family	9
6.2 Metastability requires a trap	10
6.3 A strict promotion test	10
7 Multiple rows and moving fronts	10
8 Observable diagnostics	11
9 Numerical execution and exactness	12
9.1 Boundary events need special treatment	12
9.2 What an exactness certificate would contain	13
10 Contracts for physical applications	13

11 Relation to established mathematics	14
12 Claim status and theorem ownership	15
13 Completion and falsifiability contract	15
14 Conclusion	16

1 The question in its corrected form

The original paper asked whether the edge of an admissible region has distinctive dynamics. The corrected answer is:

It can, after both the admissible region and the dynamics approaching it have been specified. The static boundary determines where a selected description first fails. The generator determines how trajectories approach that boundary. A reset rule determines what, if anything, follows the exit.

These are three different mathematical objects. Confusing them was the source of most of the earlier overclaims.

The paper therefore separates:

1. the *static ledger*: which inequalities define admissibility;
2. the *local geometry*: whether their common boundary is a facet, a corner, or a singular set;
3. the *within-chart process*: drift, diffusion, forcing, and observation horizon before first exit;
4. the *exit statistics*: committors, stopping times, active rows, and exit locations;
5. the *continuation semantics*: stop, change chart, reset deterministically, or sample a reset kernel; and
6. the *physical interpretation*: a separate map from these mathematical objects to an experiment or fundamental theory.

The static ledger and its normalized bottleneck are inherited from the capacity papers [10, 9, 11]. The explicit stop/chart/reset alternatives are inherited from the capacity-gated hybrid model [8]. This paper owns the local selection-front record, the transverse-corner description, the exact normal-diffusion calculation, and the resulting diagnostic and application contracts.

1.1 What is not claimed

No theorem below says that every admissibility boundary is physically reachable, that every first exit is irreversible, or that an inadmissible point is absent from reality. It says only that a declared description no longer has all of its certified positive reserves there. Another chart, theory, or scale may remain available.

Likewise, the word *selection* does not mean that the boundary chooses one successor. It refers to the study of a stopping or transition layer. Outcome selection begins only after a continuation law has been supplied.

2 The complete static input

2.1 A vector of signed normalized reserves

Let U be a state-and-control region and let $\mathcal{I}$ be a finite set of required rows. Each row has:

$$s_i : \mathcal{D}_i \longrightarrow \mathbb{R}, \quad \sigma_i > 0, \quad r_i = \frac{s_i}{\sigma_i}, \quad i \in \mathcal{I}.$$

Here $s_i > 0$ means that row i passes, σ_i is a declared reference scale, and the common domain is $\mathcal{D} = \bigcap_{i \in \mathcal{I}} \mathcal{D}_i$. Every row also carries provenance: its norm, tolerance, source theorem or assumption, uncertainty, and evaluation method.

The primary object is the vector

$$r(x) = (r_i(x))_{i \in \mathcal{I}}.$$

The signed bottleneck and admissible set are

$$D(x) = \min_{i \in \mathcal{I}} r_i(x), \quad \mathcal{A} = \{x \in \mathcal{D} : D(x) > 0\}. \quad (1)$$

The clipped visualization $C = \min\{1, \max\{0, D\}\}$ may be useful, but it must not replace the signed rows in an exit calculation.

For a physical application, a single spectral gap is not a complete margin. At minimum, every condition actually used by the reduced description must appear as its own row. Typical families are:

Row family	Question answered	Example signed reserve
Gap	Is the selected spectral sector separated?	$(\gamma - \gamma_{\min})/\sigma_{\text{gap}}$.
Projector or resolvent	Is the selected projection bounded and controlled?	$(K_{\max} - K)/\sigma_{\text{proj}}$.
Well-posedness	Are the operator domain, evolution, or solution hypotheses valid?	A normalized lower bound for the relevant estimate.
Descent or leakage	Does the selected sector remain closed to the declared tolerance?	$(\ell_{\max} - \ell)/\sigma_{\text{desc}}$.
Coherent stability	Is the complementary or transverse mode stable?	$(\omega_Q - \omega_{\min})/\sigma_{\text{coh}}$.
Truncation	Is the omitted remainder below the accepted error?	$(E_{\max} - E)/\sigma_{\text{tr}}$.
Numerical certificate	Does the interval, residual, or majorant test pass strictly?	A normalized certified inclusion radius.

The table is a schema, not a claim that every problem uses exactly these seven rows. Omitting a row that the downstream argument needs makes the front incomplete.

2.2 Exact boundary, front layer, and active rows

Definition 2.1 (Static selection-front layer). For a fixed normalized record and $\delta > 0$, define

$$\mathcal{F}_\delta = \{x \in \mathcal{D} : 0 < D(x) \leq \delta\}.$$

The exact admissibility boundary is

$$\partial_{\mathcal{D}}\mathcal{A} = \{x \in \mathcal{D} : D(x) = 0\}$$

when no domain boundary intervenes. The active-row set is

$$I_{\text{act}}(x) = \{i \in \mathcal{I} : r_i(x) = D(x)\}.$$

The layer thickness δ is a declared diagnostic resolution in the chosen normalized record. It is not a universal length, energy, or time. A controlled change of row normalization changes its numerical value.

2.3 Exact and numerical guards are not the same

Suppose a computation returns $\hat{r}_i(x)$ with certified errors

$$|r_i(x) - \hat{r}_i(x)| \leq e_i(x).$$

Then

$$D^-(x) = \min_i (\hat{r}_i(x) - e_i(x)) > 0$$

certifies exact admissibility. A numerical event guard may be chosen as

$$\mathcal{G}_{\text{num}} = \{x : D^-(x) \leq \delta_{\text{evt}}\},$$

where $\delta_{\text{evt}} \geq 0$ includes event-location tolerance and any declared safety reserve.

Reaching $\mathcal{G}_{\text{num}}$ need not mean that $D = 0$. It may be an early, conservative stop. Conversely, stepping across the exact boundary between two discrete time samples is possible unless event detection is included. Every numerical selection-front claim must record which guard was used.

3 Local geometry: facets, corners, and singular points

The corrected capacity-dynamics paper proves that one uniquely active C^1 row with nonzero gradient defines a local codimension-one hypersurface [11]. The next question is what happens when several rows become active together.

Proposition 3.1 (Transverse active rows give an orthant corner). *Let $U \subset \mathbb{R}^n$ be open and let $x_0 \in U$. Suppose a set $J \subset \mathcal{I}$ of $k \leq n$ rows satisfies*

$$r_j(x_0) = 0 \quad (j \in J), \quad r_i(x_0) > 0 \quad (i \notin J),$$

and the covectors $\{\text{dr}_j(x_0) : j \in J\}$ are linearly independent. Then there is a local C^1 coordinate chart

$$x \mapsto ((r_j(x))_{j \in J}, y(x)) \in \mathbb{R}^k \times \mathbb{R}^{n-k}$$

in which the admissible set is locally

$$(0, \infty)^k \times \mathbb{R}^{n-k}.$$

The common active stratum $\{r_j = 0 : j \in J\}$ has codimension k , and the local boundary is the union of its coordinate faces.

Proof. The map $r_J = (r_j)_{j \in J}$ has rank k at x_0 . The constant rank theorem completes its components to local coordinates (r_J, y) . Continuity keeps every inactive row positive after the neighborhood is reduced. The sign conditions for membership in $\mathcal{A}$ are therefore exactly $r_j > 0$ for $j \in J$, giving the stated orthant. $\square$

Remark 3.2. At a row tie, the scalar minimum $D = \min_i r_i$ is generally nondifferentiable even though every row is smooth. A force, normal coordinate, or boundary velocity should therefore be computed rowwise or within a declared stratum. Differentiating the bottleneck as though it were globally smooth is not justified.

If the active gradients lose rank, the boundary can have cusps, self-contact, higher-order tangency, or an open zero set. “A front is a smooth surface” is consequently a local theorem under regularity and transversality hypotheses, not part of the definition.

4 The additional dynamical data

4.1 A typed front record

Definition 4.1 (Dynamical selection-front record). A dynamical selection-front record consists of:

$$\mathcal{S}_{\text{front}} = (\mathcal{R}_{\text{adm}}, \{\mathcal{L}_\eta\}_{\eta \in E}, \mu_0, \mathcal{G}, \mathcal{B}, \mathcal{K}, \mathcal{O}, T_{\text{obs}}),$$

where:

1. $\mathcal{R}_{\text{adm}}$ is the complete normalized row record;
2. $\eta \in E$ is a declared model or noise parameter, and $\mathcal{L}_\eta$ is the within-chart generator;
3. μ_0 is the initial law;
4. $\mathcal{G}$ is the exact or numerical guard;
5. $\mathcal{B} \subseteq \mathcal{A}$ is an optional interior return or comparison set;
6. $\mathcal{K}$ is absent for stopping, a chart-transition map, a deterministic reset, or a Markov reset kernel;
7. $\mathcal{O}$ is the declared observable family; and
8. T_{obs} is the observation horizon.

For an Ito diffusion in local coordinates,

$$dX_t = b_\eta(X_t) dt + \sqrt{2\eta} \Sigma_\eta(X_t) dW_t, \tag{2}$$

the generator is

$$\mathcal{L}_\eta f = b_\eta \cdot \nabla f + \eta \operatorname{tr}(a_\eta \nabla^2 f), \quad a_\eta = \Sigma_\eta \Sigma_\eta^\top.$$

Regularity, nonexplosion, degeneracy, and boundary accessibility hypotheses must be stated for the chosen model. A jump process, delay equation, deterministic flow, or stochastic partial differential equation requires its own solution concept rather than being hidden in $\mathcal{L}_\eta$.

4.2 Committors and stopping times

Let

$$\tau_{\mathcal{G}} = \inf\{t \geq 0 : X_t \in \mathcal{G}\}, \quad \tau_{\mathcal{B}} = \inf\{t \geq 0 : X_t \in \mathcal{B}\}.$$

The guard committor is

$$q_{\eta}(x) = \mathbb{P}_x(\tau_{\mathcal{G}} < \tau_{\mathcal{B}}).$$

Under standard regularity assumptions it solves the boundary-value problem

$$\mathcal{L}_{\eta}q_{\eta} = 0, \quad q_{\eta}|_{\mathcal{G}} = 1, \quad q_{\eta}|_{\mathcal{B}} = 0.$$

This function is often a better operational definition of a dynamical front than the raw strip $\mathcal{F}_{\delta}$: its level sets report the actual competition between drift, diffusion, geometry, and the selected comparison sets.

The equation is not determined by D . Two models with the same admissible set and different generators can have entirely different committors and exit-time laws.

4.3 A diagnostic barrier becomes a force only when inserted

On a positive-row domain one may define

$$U_{\text{bar}}(x) = - \sum_{i \in \mathcal{I}} \alpha_i \log r_i(x), \quad \alpha_i > 0.$$

As a plotted scalar, U_{bar} is only a diagnostic. If the drift is changed to

$$b_{\eta} = b_{\eta}^{(0)} - M \nabla U_{\text{bar}},$$

then $-M \nabla U_{\text{bar}}$ is a genuine feedback force or penalty in that reduced model. The coefficients α_i and mobility M are additional constitutive inputs. The barrier does not become non-dynamical merely because its intended interpretation is “admissibility repair.”

This barrier should also not be confused with the potential in a metastable Eyring–Kramers model. They coincide only if that identification is explicitly made and the resulting drift and basin structure are verified.

4.4 Exit is not continuation

The stopped path and its exit location are determined by the within-chart process. A successor state is not. At the guard the model must choose:

$$\text{stop,} \quad \text{change chart,} \quad \text{apply a reset map,} \quad \text{sample a reset kernel.}$$

These choices produce different path laws. In particular, an exit distribution on $\mathcal{G}$ is not a distribution of post-exit outcomes unless a map or kernel from guard states to restart states has been supplied. Stochastic hybrid systems with boundary-hitting resets provide the appropriate established framework [6, 1].

5 An exactly solvable normal front

The simplest calculation already shows why no universal boundary-layer law can follow from the static margin.

5.1 Model

Let $R_t \in (0, L)$ be a signed inward normal coordinate, with $R = 0$ the guard and $R = L$ an interior comparison surface. Consider

$$dR_t = b dt + \sqrt{2\eta} dW_t, \quad \eta > 0, \quad (3)$$

stopped at

$$\tau = \tau_0 \wedge \tau_L, \quad \tau_a = \inf\{t \geq 0 : R_t = a\}.$$

Positive b points away from the guard, negative b points toward it, and $b = 0$ is unbiased diffusion.

Theorem 5.1 (Exact committor and mean exit time). *For the stopped diffusion (3), define*

$$p_\eta(x) = \mathbb{P}_x(\tau_0 < \tau_L), \quad T_\eta(x) = \mathbb{E}_x[\tau], \quad 0 < x < L.$$

If $b \neq 0$, then

$$p_\eta(x) = \frac{e^{-bx/\eta} - e^{-bL/\eta}}{1 - e^{-bL/\eta}}, \quad (4)$$

$$T_\eta(x) = \frac{L}{b} \frac{1 - e^{-bx/\eta}}{1 - e^{-bL/\eta}} - \frac{x}{b}. \quad (5)$$

If $b = 0$, then

$$p_\eta(x) = 1 - \frac{x}{L}, \quad T_\eta(x) = \frac{x(L-x)}{2\eta}. \quad (6)$$

Proof. The generator is

$$\mathcal{L}_\eta = b \frac{d}{dx} + \eta \frac{d^2}{dx^2}.$$

The committor solves

$$\mathcal{L}_\eta p_\eta = 0, \quad p_\eta(0) = 1, \quad p_\eta(L) = 0.$$

For $b \neq 0$, its general form is $A + Be^{-bx/\eta}$; imposing the boundary values gives (4). For $b = 0$, the solution is affine.

Dynkin's formula gives

$$\mathcal{L}_\eta T_\eta = -1, \quad T_\eta(0) = T_\eta(L) = 0.$$

For $b \neq 0$, integrating the ordinary differential equation twice gives

$$T_\eta(x) = -\frac{x}{b} + A + Be^{-bx/\eta}.$$

The two boundary values yield (5). When $b = 0$, solving $\eta T_\eta'' = -1$ gives (6). $\square$

5.2 A genuine boundary layer, under one sign of drift

Corollary 5.2 (Exact committor-layer thickness). *Assume $b > 0$. For any fixed $q \in (0, 1)$, the unique point $x_q \in (0, L)$ satisfying $p_\eta(x_q) = q$ is*

$$x_q = -\frac{\eta}{b} \log\left(q + (1-q)e^{-bL/\eta}\right).$$

Consequently,

$$x_q = \frac{\eta}{b} \log \frac{1}{q} + O\left(\eta e^{-bL/\eta}\right) \quad (\eta \downarrow 0).$$

Proof. Solving (4) for x gives the exact expression. For fixed $q > 0$, expand the logarithm around q as $e^{-bL/\eta} \rightarrow 0$. $\square$

Thus a thin exponential committor layer is real in this model, and its scale is η/b . But it is not universal:

- If $b > 0$, $p_\eta(x)$ is exponentially small away from the guard, while $T_\eta(x) \rightarrow (L - x)/b$.
- If $b = 0$, the committor is linear throughout the interval and $T_\eta(x)$ scales as η^{-1} .
- If $b < 0$, the process follows the deterministic drift toward the guard and $T_\eta(x) \rightarrow x/|b|$; on compact subsets away from L , $p_\eta(x) \rightarrow 1$.

The same static interval therefore supports a thin exponential layer, a domain-wide linear profile, or almost-certain guard hitting. It supports no universal sigmoid and no automatic exponentially long metastable time.

5.3 What this model does and does not establish

The normal model is a local reduced calculation. It becomes an approximation to a higher-dimensional front only after:

1. a regular active row provides a normal coordinate;
2. tangential motion and curvature terms are bounded or retained;
3. the projected drift and diffusion are derived with an error bound;
4. the approximation remains valid over the stopping-time scale of interest; and
5. the comparison surface $R = L$ is chosen independently of the desired result.

Without those items, fitting b and η to observed exit data is a phenomenological model, not a derivation from MTT geometry.

6 When large deviations and metastability are valid

6.1 Large deviations require a family

Large-deviation theory concerns a declared family such as

$$dX_t^\eta = f(X_t^\eta) dt + \sqrt{2\eta} \Sigma(X_t^\eta) dW_t, \quad \eta \downarrow 0.$$

Under suitable regularity and nondegeneracy assumptions, paths have an action functional. With $a = \Sigma \Sigma^\top$ invertible, one common normalization is

$$I_{0T}(\phi) = \frac{1}{4} \int_0^T (\dot{\phi} - f(\phi))^\top a(\phi)^{-1} (\dot{\phi} - f(\phi)) dt.$$

Path probabilities then have logarithmic asymptotics governed by $\inf I_{0T}/\eta$, subject to the standard open/closed set qualifications [4, 7].

One simulation at one noise amplitude does not establish a large-deviation principle. Neither does a small reserve D . The small-noise parameter, topology on path space, action, and limiting family must all be supplied.

6.2 Metastability requires a trap

Metastability is stronger than proximity to a guard. It requires a long-lived region, a shorter local relaxation scale, and a much longer escape scale. For the standard reversible gradient diffusion

$$dX_t = -\nabla V(X_t) dt + \sqrt{2\eta} dW_t, \quad (7)$$

suppose m is a nondegenerate local minimum and z is the relevant nondegenerate index-one saddle. In the standard ordered-well setting, the Eyring–Kramers transition law has the form

$$\mathbb{E}_m \tau \sim \frac{2\pi}{|\lambda_-(z)|} \sqrt{\frac{|\det \nabla^2 V(z)|}{\det \nabla^2 V(m)}} \exp\left(\frac{V(z) - V(m)}{\eta}\right), \quad \eta \downarrow 0, \quad (8)$$

where $\lambda_-(z)$ is the unique negative Hessian eigenvalue. Rigorous versions require precise basin, saddle, regularity, and ordering hypotheses [3, 2].

Equation (8) is powerful precisely because its hypotheses are restrictive. Degenerate saddles, non-gradient drift, multiple equal escape routes, state-dependent or degenerate noise, non-Markov memory, moving guards, and hybrid resets can alter the prefactor, the exponent, or the whole asymptotic form.

6.3 A strict promotion test

A selection-front study may use the words *metastable* or *Kramers* only after it records:

1. the process family and small parameter;
2. an attracting set or local equilibrium;
3. a basin and exit set;
4. a quasipotential or actual potential barrier;
5. the relevant regularity and nondegeneracy conditions;
6. the initial or quasistationary law;
7. the asymptotic observable and error term; and
8. whether a reset changes the stopped process.

The v1 paper supplied none of these as a common cross-domain record. Its universal metastability claim is therefore withdrawn, not merely renamed.

7 Multiple rows and moving fronts

If one row $r_j(t, x)$ is uniquely active and regular, its zero set has normal velocity

$$V_n = -\frac{\partial_t r_j}{|\nabla r_j|}.$$

If it obeys

$$\partial_t r_j + v \cdot \nabla r_j = s_j,$$

then

$$V_n = v \cdot n - \frac{s_j}{|\nabla r_j|}.$$

These are standard level-set identities, already derived in the corrected capacity-dynamics paper and situated in the level-set literature [11, 12]. They track a front after its row evolution has been supplied; they do not select that evolution.

At a transverse corner, several normal coordinates compete. The appropriate local object is the vector

$$R_J(X_t) = (r_j(X_t))_{j \in J},$$

not one differentiable scalar minimum. Ito's formula gives each row its own projected drift, quadratic variation, and cross-variation. The probability of first reaching a particular face then depends on the full covariance matrix and tangential dynamics. Even the identity of the first failing row is generator-dependent.

If the rows themselves depend on time, the admissible set moves. The stopping problem must then be posed in space-time, and a row can reach zero because the state moves, the tolerance changes, or both. These mechanisms should not be merged into a single claim that “capacity was consumed.”

8 Observable diagnostics

A front claim should be attached to observables that can be computed, bounded, or measured. Useful examples are:

$$\begin{aligned}\tau_{\mathcal{G}} &= \inf\{t : X_t \in \mathcal{G}\}, \\ q(x) &= \mathbb{P}_x(\tau_{\mathcal{G}} < \tau_{\mathcal{B}}), \\ \Theta_{\delta}(T) &= \frac{1}{T} \int_0^T \mathbf{1}_{\{0 < D(X_t) \leq \delta\}} dt, \\ \nu_{\text{exit}}(A) &= \mathbb{P}_{\mu_0}(X_{\tau_{\mathcal{G}}} \in A), \\ \pi_i &= \mathbb{P}_{\mu_0}(i \in I_{\text{act}}(X_{\tau_{\mathcal{G}}}).\end{aligned}$$

Proposed label	Minimum operational test	What it does not prove
Thin boundary layer	Committer level-set widths shrink under a declared asymptotic family.	A new phase or force.
Metastability	Separation between local relaxation and exit times, with a quasistationary or basin analysis.	Universal Kramers scaling.
Protocol sensitivity	Exit laws change under predeclared variations of initial law, forcing, guard, or reset.	Fundamental indeterminism.
Active-row switching	The identity or multiplicity of limiting rows changes along paths.	Nonsmooth physical motion.
Abrupt transition	A stated observable develops a controlled sharp asymptotic or discontinuity.	Breakdown of every alternative chart.
Irreversibility	Forward and reversed path measures, records, or semigroups fail a declared reversibility criterion.	Irreversibility from noninjectivity alone.

Words such as *hesitation*, *channeling*, or *selection* can remain useful descriptive labels, but the archival claim should name the observable, parameter family, comparison baseline, confidence or error bound, and acceptance criterion.

9 Numerical execution and exactness

9.1 Boundary events need special treatment

A fixed-step Euler scheme can move from $D > 0$ to $D < 0$ without recording the crossing. For killed diffusions, boundary hitting can reduce the weak convergence order of a naive discrete scheme [5]. A credible numerical front calculation should:

1. retain every signed row rather than only the clipped bottleneck;
2. bracket the first row crossing within a step;
3. refine or interpolate the event using a stated method;
4. distinguish the exact guard from a numerical safety guard;
5. stop before applying a separately declared reset;
6. record random seeds and all constitutive parameters;
7. repeat across time-step and guard tolerances; and
8. report sampling error separately from discretization and model error.

9.2 What an exactness certificate would contain

For a theorem-level computational claim, the archival packet should contain:

1. source hashes for the row functions and generator;
2. interval or otherwise certified row-evaluation errors;
3. a verified positive reserve before each accepted step;
4. a certified event bracket or analytic exit solution;
5. convergence bounds for the chosen stopped-process observable;
6. the post-exit semantics and reset-support check, if any; and
7. a verifier independent enough to reject altered rows, parameters, or outputs.

The present paper contains an analytic calculation, not a numerical result packet. No empirical or physical prediction is promoted.

10 Contracts for physical applications

The earlier paper moved directly from a formal boundary to cosmology, black holes, measurement, and limits on coherence. The corrected route is to state what each application would have to add.

Application	Required source data	Status here
Vacuum or compactification selection	A selected configuration space, complete geometric and operator rows, a branch dynamics, noise or control law, and a map from exit/reset states to four-dimensional observables.	Not derived.
EFT or RG validity	A rigorous truncation/error row, well-posed flow, regulator and scheme map, and a comparison to an overlapping description.	A front can mark first certificate failure only.
Cosmological initial data	A state space, measure or deterministic law, constraint surface, observation map, and a justified extrapolation domain.	No global measure conclusion.
Black-hole dynamics	Semiclassical validity rows, a spacetime and quantum-state model, horizon observables, and a continuation law past first failure.	No horizon, information-loss, or evaporation theorem.
Quantum measurement	A system–apparatus algebra, state, interaction, instrument or detector law, outcome record, and probability theorem.	No collapse or Born-rule derivation.
Scalable coherence	An operational coherence measure, noise and control model, device geometry, and asymptotic scaling family.	No universal coherence or entanglement-depth bound.

These are not rhetorical disclaimers. They are the missing maps that would turn a conditional first-exit model into a claim about nature.

11 Relation to established mathematics

The mathematical neighbors are direct:

- first-exit probabilities and singularly perturbed boundary-value problems [7];
- Freidlin–Wentzell large deviations for small random perturbations [4];
- potential-theoretic metastability and rigorous Eyring–Kramers laws [3, 2];
- hybrid systems and boundary-hitting resets [6, 1];
- level-set motion for supplied moving-front equations [12]; and
- numerical approximation of killed diffusions [5].

The MTT-specific contribution is not the invention of exit theory. It is the insistence that the exit domain be built from a complete, provenance-bearing vector of normalized admissibility rows, and that static failure, within-chart motion, numerical stopping, and post-exit selection remain separate typed layers.

12 Claim status and theorem ownership

Statement	Status	Reason
A complete row record defines an exact first-failure set	Inherited exact fact	Owned by the normalized capacity paper.
A unique regular row defines a local hypersurface	Inherited conditional theorem	Owned by the capacity-dynamics paper.
Transverse simultaneous rows form a local orthant corner	Proved here	Proposition 3.1.
The normal constant-drift diffusion has the stated committor and exit time	Proved here	Theorem 5.1.
Its outward-drift committor has scale η/b	Proved here	Corollary 5.2.
Every admissibility boundary has a thin layer	False without dynamics	The exact model gives three different regimes.
Every front is metastable or obeys Kramers law	Withdrawn	Requires an attracting basin, barrier, small-noise family, and regular saddle data.
Exit selects a unique successor	Withdrawn	A reset map or kernel is independent data.
Selection fronts explain collapse, horizons, or cosmological measures	Open application proposals	The physical source maps listed in Section 10 are absent.

This paper is therefore a conditional mathematical reconstruction at the level of a reduced first-exit model. It is not a selected-source theorem for a specific MTT compactification, Standard Model sector, quantum measurement apparatus, or quantum-gravity observable.

13 Completion and falsifiability contract

A proposed physical selection front is complete only if another researcher can answer all of the following from the archive:

1. What is the state-and-control space?
2. What are all required signed rows, scales, norms, and provenances?
3. Which boundary point or stratum is under study?
4. Is the geometry regular, transverse, or singular?
5. What is the within-chart generator or deterministic evolution?
6. What parameter tends to zero, infinity, or a threshold?
7. What are the initial law, guard, comparison set, and observation horizon?

8. Which committor, exit time, exit law, or active-row statistic is claimed?
9. What are the analytic, numerical, and sampling errors?
10. Does the process stop, change chart, or reset after exit?
11. Which physical observable does the mathematical output represent?
12. Which held-out data could contradict the construction?

The selection-front interpretation fails for a proposed application if:

- no complete common row record can be formed;
- the claimed front disappears under allowed normalization or certification refinement;
- the observed exit law disagrees with the supplied generator outside the declared error;
- a purported universal scaling changes when an unrecorded drift, covariance, guard, or reset is exposed;
- the physical observable map is nonunique and no source selects one; or
- an overlapping effective description remains controlled across the alleged terminal boundary.

14 Conclusion

An admissibility boundary is a precise and useful object: it is the first place where one or more declared control rows lose positive reserve. A selection front becomes dynamical only after a process is supplied.

The exact normal calculation makes the central lesson unavoidable. The same static interval exhibits an exponential committor layer, a linear profile, or almost-certain guard hitting depending only on the drift. Metastability and Eyring–Kramers scaling are valid and important in their own basin-and-barrier regime, but they are not consequences of a small bottleneck reserve.

The corrected concept is therefore narrower than Version 1 and more useful. It provides a disciplined bridge from MTT admissibility records to first-exit analysis while keeping geometry, dynamics, numerics, reset, and physical interpretation separate. That separation is what makes future applications testable.

Reproducibility statement

All formulas in Section 5 follow from the two elementary boundary value problems

$$bp' + \eta p'' = 0, \quad bT' + \eta T'' = -1$$

with the displayed boundary conditions. The paper contains no fitted parameter, simulation output, or mapped numerical-result packet. The canonical source, generated Markdown, bibliography, revision audit, and release PDF are maintained in the MTT papers repository. The public repository identifiers are included in the release metadata.

References

- [1] Julien Bect, Hana Baili, and Gilles Fleury. Fokker–planck–kolmogorov equation for stochastic differential equations with boundary hitting resets, 2005.
- [2] Nils Berglund. Kramers’ law: Validity, derivations and generalisations, 2011.
- [3] Anton Bovier, Michael Eckhoff, Veronique Gayraud, and Markus Klein. Metastability in reversible diffusion processes i: Sharp asymptotics for capacities and exit times. *Journal of the European Mathematical Society*, 6(4):399–424, 2004.
- [4] Mark I. Freidlin and Alexander D. Wentzell. *Random Perturbations of Dynamical Systems*, volume 260 of *Grundlehren der mathematischen Wissenschaften*. Springer, 3 edition, 2012.
- [5] Emmanuel Gobet. Weak approximation of killed diffusion using euler schemes. *Stochastic Processes and their Applications*, 87(2):167–197, 2000.
- [6] Rafal Goebel, Ricardo G. Sanfelice, and Andrew R. Teel. *Hybrid Dynamical Systems: Modeling, Stability, and Robustness*. Princeton University Press, 2012.
- [7] Bernard J. Matkowsky and Zeev Schuss. The exit problem for randomly perturbed dynamical systems. *SIAM Journal on Applied Mathematics*, 33(2):365–382, 1977.
- [8] Peter Nero. Capacity-gated projection dynamics: A typed hybrid stochastic model and executable specification, 2026.
- [9] Peter Nero. Coherence capacity as a control budget for effective descriptions: Robustness, composition, and the limits of resource language, 2026.
- [10] Peter Nero. Coherence capacity in modal triplet theory: Normalized margins, metric clearance, and the invariance class, 2026.
- [11] Peter Nero. Dynamics of coherence capacity: Constitutive transport, balance laws, and first exit, 2026.
- [12] Stanley Osher and James A. Sethian. Fronts propagating with curvature-dependent speed: Algorithms based on hamilton–jacobi formulations. *Journal of Computational Physics*, 79(1):12–49, 1988.